

Faculty of Computing

Riphah International University, Islamabad

Data Structures



Assignment # 01

Submitted by

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to

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on

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1-) Project Description:-

The FinTech Trading Management System is a Java based application that simulates stock market trading and management. It allows users to view market assets, search and sort stock data, manage a custom watchlist, queue buy and sell orders, track portfolio and maintain a history of transaction orders.

2-) Data Structures and Algorithms Used:-

- **Arrays:-**

Used to store the list of market assets in Main.java, owned assets and quantities in Portfolio.java, and as the underlying container in TransactionStack and TradingOrderQueue.

- **Linear Search:-**

Used in linearSearch() in Main.java to locate an asset by checking each array element sequentially by ID, as well as in findAssetById() and Portfolio.

- **Binary Search:-**

Used in binarySearch() in Main.java to quickly find an asset by ID by repeatedly dividing the sorted array in half.

- **Bubble Sort:-**

Used in `bubbleSortByPrice()` in `Main.java` to sort market assets by price in ascending order using adjacent element comparisons.

- **Selection Sort:-**

Used in `selectionSortByChangeDesc()` in `Main.java` to find and place assets in descending order based on their 24-hour price change percentage.

- **Insertion Sort:-**

Used in `insertionSortById()` in `Main.java` to build a sorted array of assets by Asset ID in ascending order.

- **Merge Sort:-**

Used in `mergeSort()` in `Main.java` to divide the market assets array into sub-arrays and merge them back together sorted by price in ascending order.

- **Quick Sort:-**

Used in `quickSort()` in `Main.java` to partition the asset array around a pivot and sort items by Asset ID in ascending order.

DATE: ___/___/___

- **Linked List:-**

Used in Watchlist.java via a custom singly linked list to dynamically add, remove, search, and view watchlisted assets.

- **Recursion:-**

Used in Portfolio.java inside displayPortfolioRecursive() to print holdings and calculate TotalValueRecursive() to sum up total portfolio value without standard loops

- **Stack:-**

Used in TransactionStack.java as a LIFO array structure to store completed trades and pop the most recent transaction

- **Queue:-**

Used in TradingOrderQueue.java as a FIFO array structure to manage incoming buy/sell orders in the order they arrive until they are processed.

3-) Three Dry Runs:-

1-) Linear Search:-

Target ID = 103.

Searching:-

- $i = 0 : arr[0] = 101 \rightarrow 101 \neq 103$ (continue)

- $i = 1$: $arr[1] = 102 \rightarrow 102 \neq 103$ (Continue).
- $i = 2$: $arr[2] = 103 \rightarrow 103 = 103$ (ID Found).

Result:-

Element found at index 2.

2-) Stack (Push & Pop):-

Initial:- $top = -1$, $stack = (null, null)$.

- $Push(Transaction 1)$:
 top increases to 0.
 $stack[0] = Transaction 1$.
 - $Push(Transaction 2)$:
 top increases to 1.
 $stack[1] = Transaction 2$.
 - $Pop()$:-
 returns $stack[1]$ (Transaction 2)
 $stack[1] = null$, top decreases to 0.
- Result:- Pops Transaction 2 (LIFO order).

3-) Queue:- (Queue & Dequeue)

Initial:- $front = 0$, $rear = -1$, $queue = (null, null)$.

- $Enqueue(Order 1)$:-
 $rear$ increases to 0.
 $queue[0] = Order 1$.
- $Enqueue(Order 2)$:-
 $rear$ increases to 1.
 $queue[1] = Order 2$.
- $Dequeue()$:-
 returns $queue[0]$ (Order 1)
 $queue[0] = null$
 $front$ increases to 1.

Result:- Dequeues Order 1 (FIFO order)

Sample Output:

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=====
                        FINTECH TRADING MANAGEMENT SYSTEM
=====

1. Manage Market Assets
2. Search Asset
3. Sort Market Assets
4. Manage Watchlist
5. Calculate Portfolio Value
6. Manage Transactions
7. Manage Trading Orders
0. Exit
Enter your choice:
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-----
                        MARKET ASSETS
-----

1. Display All Assets
2. Calculate Average Price
3. Find Highest-Priced Asset
4. Find Lowest-Priced Asset
0. Back to Main Menu
Enter your choice: 1

--- MARKET ASSETS ---
ID: 101 | Name: Engro Corporation      | Type: Conglomerate | Price: Rs. 345.50 | 24h Change: +2.15%
ID: 102 | Name: Lucky Cement              | Type: Cement       | Price: Rs. 810.00 | 24h Change: +1.40%
ID: 103 | Name: Hub Power Co. (HUBC)       | Type: Power        | Price: Rs. 125.75 | 24h Change: -0.85%
ID: 104 | Name: Habib Bank Ltd (HBL)       | Type: Banking      | Price: Rs. 138.20 | 24h Change: +1.10%
ID: 105 | Name: Oil & Gas Dev (OGDC)       | Type: Energy       | Price: Rs. 118.90 | 24h Change: -1.50%
ID: 106 | Name: Systems Limited            | Type: Technology   | Price: Rs. 412.30 | 24h Change: +3.45%
ID: 107 | Name: Pakistan State Oil         | Type: Energy       | Price: Rs. 195.40 | 24h Change: +0.55%
ID: 108 | Name: Fauji Fertilizer           | Type: Agriculture  | Price: Rs. 168.10 | 24h Change: +0.90%
ID: 109 | Name: MCB Bank Limited           | Type: Banking      | Price: Rs. 210.60 | 24h Change: +1.75%
ID: 110 | Name: TRG Pakistan               | Type: Technology   | Price: Rs. 45.25  | 24h Change: -4.20%

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                        MARKET ASSETS
-----

1. Display All Assets
2. Calculate Average Price
3. Find Highest-Priced Asset
4. Find Lowest-Priced Asset
0. Back to Main Menu
Enter your choice: 2
Average Asset Price: Rs. 257.00
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MARKET ASSETS
-----
1. Display All Assets
2. Calculate Average Price
3. Find Highest-Priced Asset
4. Find Lowest-Priced Asset
0. Back to Main Menu
Enter your choice: 3
Highest-Priced Asset: ID: 102 | Name: Lucky Cement | Type: Cement | Price: Rs. 810.00 | 24h Change: +1.40%
-----
MARKET ASSETS
-----
1. Display All Assets
2. Calculate Average Price
3. Find Highest-Priced Asset
4. Find Lowest-Priced Asset
0. Back to Main Menu
Enter your choice: 4
Lowest-Priced Asset: ID: 110 | Name: TRG Pakistan | Type: Technology | Price: Rs. 45.25 | 24h Change: -4.20%

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SEARCH ASSET
-----
1. Linear Search
2. Binary Search
0. Back to Main Menu
Enter your choice: 1
Enter Asset ID to search: 104

--- Executing Linear Search for ID: 104 ---
Checking index 0 [ID: 101]...
Checking index 1 [ID: 102]...
Checking index 2 [ID: 103]...
Checking index 3 [ID: 104]...
FOUND: ID: 104 | Name: Habib Bank Ltd (HBL) | Type: Banking | Price: Rs. 138.20 | 24h Change: +1.10%
-----
SEARCH ASSET
-----
1. Linear Search
2. Binary Search
0. Back to Main Menu
Enter your choice: 2
Enter Asset ID to search: 103

--- Executing Binary Search for ID: 103 ---
FOUND: ID: 103 | Name: Hub Power Co. (HUBC) | Type: Power | Price: Rs. 125.75 | 24h Change: -0.85%

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                        SORT MARKET ASSETS
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1. Bubble Sort (by Price - Ascending)
2. Selection Sort (by Change % - Descending)
3. Insertion Sort (by Asset ID - Ascending)
4. Merge Sort (by Price - Ascending)
5. Quick Sort (by Asset ID - Ascending)
0. Back to Main Menu
Enter your choice: 1

[Bubble Sort] Sorted by Price:
ID: 110 | Name: TRG Pakistan          | Type: Technology   | Price: Rs. 45.25   | 24h Change: -4.20%
ID: 105 | Name: Oil & Gas Dev (OGDC)       | Type: Energy       | Price: Rs. 118.90  | 24h Change: -1.50%
ID: 103 | Name: Hub Power Co. (HUBC)        | Type: Power        | Price: Rs. 125.75  | 24h Change: -0.85%
ID: 104 | Name: Habib Bank Ltd (HBL)        | Type: Banking      | Price: Rs. 138.20  | 24h Change: +1.10%
ID: 108 | Name: Fauji Fertilizer            | Type: Agriculture  | Price: Rs. 168.10  | 24h Change: +0.90%
ID: 107 | Name: Pakistan State Oil          | Type: Energy       | Price: Rs. 195.40  | 24h Change: +0.55%
ID: 109 | Name: MCB Bank Limited            | Type: Banking      | Price: Rs. 210.60  | 24h Change: +1.75%
ID: 101 | Name: Engro Corporation           | Type: Conglomerate | Price: Rs. 345.50  | 24h Change: +2.15%
ID: 106 | Name: Systems Limited            | Type: Technology   | Price: Rs. 412.30  | 24h Change: +3.45%
ID: 102 | Name: Lucky Cement               | Type: Cement       | Price: Rs. 810.00  | 24h Change: +1.40%

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                        MANAGE WATCHLIST
-----

1. Add Asset to Watchlist (Insert)
2. Remove Asset from Watchlist (Delete)
3. Search Asset in Watchlist
4. Display Watchlist
0. Back to Main Menu
Enter your choice: 1
Enter Asset ID to insert into Watchlist: 104
Added to Watchlist: Habib Bank Ltd (HBL)

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                        MANAGE WATCHLIST
-----

1. Add Asset to Watchlist (Insert)
2. Remove Asset from Watchlist (Delete)
3. Search Asset in Watchlist
4. Display Watchlist
0. Back to Main Menu
Enter your choice: 4

--- INVESTOR WATCHLIST ---
ID: 104 | Name: Habib Bank Ltd (HBL) | Type: Banking | Price: Rs. 138.20 | 24h Change: +1.10%

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=====
MANAGE WATCHLIST
=====
1. Add Asset to Watchlist (Insert)
2. Remove Asset from Watchlist (Delete)
3. Search Asset in Watchlist
4. Display Watchlist
0. Back to Main Menu
Enter your choice: 3
Enter Asset ID to search in Watchlist: 104
Found in Watchlist: ID: 104 | Name: Habib Bank Ltd (HBL) | Type: Banking | Price: Rs. 138.20 | 24h Change: +1.10%

=====
MANAGE WATCHLIST
=====
1. Add Asset to Watchlist (Insert)
2. Remove Asset from Watchlist (Delete)
3. Search Asset in Watchlist
4. Display Watchlist
0. Back to Main Menu
Enter your choice: 2
Enter Asset ID to remove: 104
Removed: Habib Bank Ltd (HBL)

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FINTECH TRADING MANAGEMENT SYSTEM
=====
1. Manage Market Assets
2. Search Asset
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4. Manage Watchlist
5. Calculate Portfolio Value
6. Manage Transactions
7. Manage Trading Orders
0. Exit
Enter your choice: 5

=====
PORTFOLIO VALUE (RECURSION)
=====
Portfolio Items:
- Engro Corporation | Quantity: 15 | Unit Price: Rs. 345.50 | Item Total: Rs. 5182.50
- Lucky Cement | Quantity: 2 | Unit Price: Rs. 810.00 | Item Total: Rs. 1620.00
- Oil & Gas Dev (OGDC) | Quantity: 10 | Unit Price: Rs. 118.90 | Item Total: Rs. 1189.00
- Systems Limited | Quantity: 5 | Unit Price: Rs. 412.30 | Item Total: Rs. 2061.50

>>> Total Calculated Portfolio Value: Rs. 10053.00 <<<

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TRADING ORDERS (QUEUE - FIFO)
-----
1. Enqueue New Trading Order
2. Dequeue / Process Order
3. Peek Front Order
4. Display All Pending Orders
0. Back to Main Menu
Enter your choice: 1
Enter Asset ID: 103
Enter Operation (BUY/SELL): buy
Enter Quantity: 10
[QUEUE ENQUEUE] Placed Order: Asset ID: 103 | Operation: BUY | Quantity: 10

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TRADING ORDERS (QUEUE - FIFO)
-----
1. Enqueue New Trading Order
2. Dequeue / Process Order
3. Peek Front Order
4. Display All Pending Orders
0. Back to Main Menu
Enter your choice: 2
[PORTFOLIO UPDATE] Added 10 unit(s) of Hub Power Co. (HUBC)
[STACK PUSH] Recorded: Tx #1001 | BUY | Hub Power Co. (HUBC) | Qty: 10 | Price: Rs. 125.75 | Total: Rs. 1257.50
Order processed and successfully recorded into Transaction Stack!

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TRANSACTIONS (STACK - LIFO)
-----
1. Pop Recent Transaction
2. Peek Most Recent Transaction
3. Display All Recent Transactions
0. Back to Main Menu
Enter your choice: 2
Top of Stack (Most Recent): Tx #1001 | BUY | Hub Power Co. (HUBC) | Qty: 10 | Price: Rs. 125.75 | Total: Rs. 1257.50

-----
TRANSACTIONS (STACK - LIFO)
-----
1. Pop Recent Transaction
2. Peek Most Recent Transaction
3. Display All Recent Transactions
0. Back to Main Menu
Enter your choice: 3

--- RECENT TRANSACTIONS (LIFO Order) ---
1. Tx #1001 | BUY | Hub Power Co. (HUBC) | Qty: 10 | Price: Rs. 125.75 | Total: Rs. 1257.50

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TRANSACTIONS (STACK - LIFO)
-----
1. Pop Recent Transaction
2. Peek Most Recent Transaction
3. Display All Recent Transactions
0. Back to Main Menu
Enter your choice: 1
[STACK POP] Removed Most Recent: Tx #1001 | BUY | Hub Power Co. (HUBC) | Qty: 10 | Price: Rs. 125.75 | Total: Rs. 1257.50

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--- The End ---